Homework 5.11. 2026

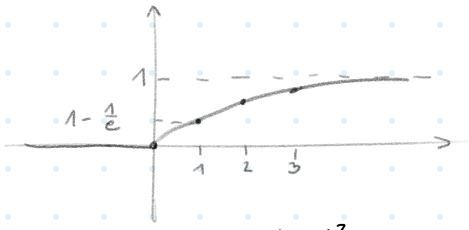
↳ exercises 4.1, 4.2, 4.4

4.1 Which of the functions are valid CDFs? Find their densities

(a) $F(x) = \begin{cases} 1 - e^{-x^2} ; & x \geq 0 \\ 0 ; & x < 0 \end{cases}$

$F(2) = 1 - e^{-4} = 1 - \frac{1}{e^4}$

$F(3) = 1 - e^{-9} = 1 - \frac{1}{e^9}$



Right continuity for $x=0$: $\lim_{\varepsilon \searrow 0} (0+\varepsilon) = \lim_{\varepsilon \searrow 0} (1 - e^{-(0+\varepsilon)^2}) = 1 - \lim_{\varepsilon \searrow 0} e^{-\varepsilon^2} = 1 - 1 = 0$

Is valid. ✓

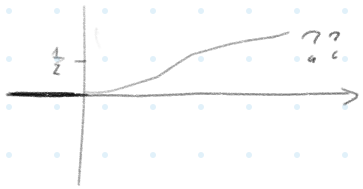
Density: differentiate $F(x)$: $F'(x) = \begin{cases} 2x\, e^{-x^2} ; & x \geq 0 \\ 0 ; & x < 0 \end{cases}$

(b) $F(x) = \begin{cases} e^{-\frac{1}{x}} ; & x > 0 \\ 0 ; & x \leq 0 \end{cases}$

$F(1) = e^{-1} = \frac{1}{e}$

$F(2) = e^{-\frac{1}{2}} = \frac{1}{e^{1/2}} = \frac{1}{\sqrt{e}}$

$F(3) = e^{-\frac{1}{3}} = \frac{1}{e^{1/3}} = \frac{1}{\sqrt[3]{e}}$



IDK why $\lim_{x \to \infty} (e^{-\frac{1}{x}}) = 1$

(c) $F(x) = \begin{cases} 0 ; & x \leq 0 \\ 1/3 ; & 0 < x \leq \frac{1}{2} \\ 1 ; & x > \frac{1}{2} \end{cases}$

Continuity in $x = \frac{1}{2}$?

$F(1/2) = \frac{1}{3}$

$\lim_{\varepsilon \searrow 0} F(1/2 + \varepsilon) = 1 \neq 1/3$

So no, not a valid CDF

4.2 Let X be a random variable with CDF $F(x) = \begin{cases} 0 ; & x < 0 \\ x^2/2 ; & 0 \leq x < 1 \\ 1/2 + p/2 ; & 1 \leq x < 2 \\ 1/2 + p/2 + \frac{1-p}{2} ; & x \geq 2 \end{cases}$

Find the prob. density/mass of X.

Differentiate: $F'(x) = \begin{cases} 0 ; & x < 0 \text{ v } x \geq 1 \\ x ; & 0 \leq x < 1 \end{cases}$

∀ discrete part: Two steps, 1 with $p = \frac{p}{2}$, 2 with $p = \frac{1-p}{2}$ — Bernoulli distrib.

$P(X = x) = \left(\frac{p}{2}\right)^{2-x} \left(\frac{1-p}{2}\right)^{x-1}$

↳ od tje to pride ??????

4.4 Let X_k, $k = 1, \ldots, n$ be rand. vars. with the Bernoulli measure as the PMF

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